

2. A student is given wire cutters and a reel of metal wire, and asked to find the resistivity of the metal. She cuts off a piece of the wire and makes measurements on it. She takes repeat readings and obtains the same values each time.

Quantity	Measurement	Instrument	Resolution of instrument
length	812 mm	metre ruler	1 mm
diameter	0.48 mm	digital calipers	0.01 mm
resistance	2.2 Ω	digital multimeter	0.1 Ω

(a) (i) Calculate the resistivity of the metal of the wire. [3]

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(ii) Calculate the **absolute** uncertainty in the resistivity giving your value to an appropriate number of significant figures. [3]

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(b) Suggest **one** way in which the student could reduce the uncertainty in her value for the resistivity, using the same reel of wire and the same instruments as before. Explain briefly why the uncertainty would be reduced. [2]

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